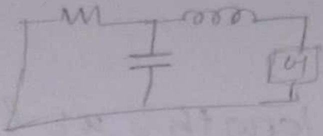
16/6/26

Transmission Line theory.

R, L, C, G → parameters in equivalent ckt of transmission.

series Parallel.

capacitance connected between resistor and inductor.



→ These four constant parameters are <u>primary constants.</u> (R, L, C, G)

Two impedance
- → series impedance
- → shunt Admittance

<u>eqn for series imp</u>

$Z = R + j\omega L$

const varying

<u>eqn for shunt</u> imp admittance

$Y = G + j\omega C$

const varying

<u>secondary constant of a Transmission line</u>

→ Propagation const $\gamma = \sqrt{ZY}$

→ characteristic impedance $\boxed{\sqrt{\frac{Z}{Y}} = Z_0}$

series imp w.r.t shunt admittance

if Perfectly matched imp then equal to Z_0

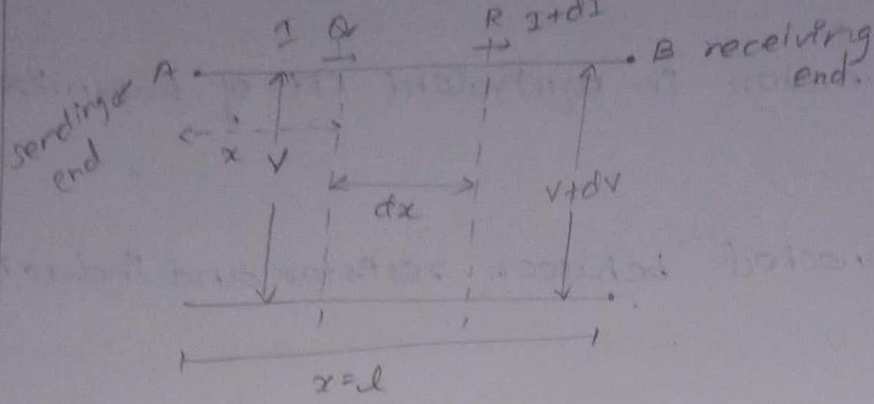
$\boxed{Z_0 = Z_{in} = Z_L}$

i/p imp

Load o/p imp

for max power transfer and zero sig. reflection

General solution of transmission line eqn:



→Let, the line AB be for length, $x = l$. and primary constant of line the R, L, C and G per km.

→ This assume that they donot vary with frequency

→consider a small section of the line Q and R of length small dx at a distance x from the sending end A

→ By making dx very small the current may be consider as a constant for voltage calculation and voltage is constant for current calculation.

→ At a point of 'Q' voltage is V and current is I.

→At a point of 'R' voltage is $V+dV$ and current is $I+dI$.

→ series impedance of small section dx will be $z\,dx$ and ||ly shunt admittance for small section is $y\,dx$.

Potential difference between a and R.

$V = IR$ make current is constant

$$v-(v+dv) = I\,z\,dx$$

$$\therefore -\frac{dv}{dx} = Iz \longrightarrow (1)$$

current between a and R

$$I = \frac{V}{R} \quad , \text{voltage is constant}$$

$$I-(I+dI) = v\,y\,dx$$

$$-dI = vy\,dx$$

$$-\frac{dI}{dx} = vy \longrightarrow (2)$$

~~diff (1) w.r.t 'x'~~

$$\cancel{-\frac{d^2v}{dx^2} = 0}$$

~~diff (2) w.r.t 'x'~~

$$\cancel{-\frac{d^2I}{dx^2} = 0}$$

diff (1) w.r.t 'x'

$$-\frac{d^2v}{dx^2} = \frac{dI}{dx}\cdot z$$

sub, $\frac{dI}{dx} = -vy$

$$\frac{d^2v}{dx^2} = vyz \longrightarrow (3)$$

$p = \sqrt{zy} \quad p^2 = zy$

$p^2 v$

diff (2) w.r.t 'x'

$$-\frac{d^2I}{dx^2} = \frac{dv}{dx}\cdot y$$

sub $\frac{dv}{dx} = -Iz$

$$\frac{d^2I}{dx^2} = Izy \longrightarrow (4)$$

Ip^2

→ eqn (3) & (4) are the differential equations of the transmission line

Fundamental of ckt of distributed constants

these equations are standard linear differential eqns with const coefficients whose solutions are,

$V = ae^{px} + be^{-px}$

incident wave ↳ reflected wave

a, b, c, d → constants

$I = ce^{px} + de^{-px}$.

from eqn ③,

$$\frac{d^2V}{dx^2} = p^2V. \longrightarrow ⑤$$

Ily, from eqn ④

$$\frac{d^2I}{dx^2} = p^2I. \longrightarrow ⑥$$

17/6/26 General equ for total voltage & current

$V = ae^{px} + be^{-px} \longrightarrow ⑦$

$I = ce^{px} + de^{-px} \longrightarrow ⑧$

The solution of the exponential are replaced by hyperbolic function

$$\cosh px = \frac{e^{px} + e^{-px}}{2}$$

Normal differ

$\cos\theta = -\sin\theta$

$$\sinh px = \frac{e^{px} - e^{-px}}{2}$$

hyperbolic differ

$\cos\theta = \sin\theta$

$(\cosh px + \sinh px)$

$e^{px} = \cosh(px) + \sinh(px)$

$(\cosh px - \sinh px)$

$e^{-px} = \cosh(px) - \sinh(px)$

sub e^{px} and e^{-px} sub in ⑦ & ⑧

$V = a[\cosh(px) + \sinh(px)] + b[\cosh(px) - \sinh(px)]$

$= a\cosh(px) + a\sinh(px) + b\cosh(px) - b\sinh(px)$

$V = \cosh(px)[a+b] + \sinh(px)[a-b]$.

$$I = c[\cosh(px) + \sinh(px)] + d[\cosh(px) - \sinh(px)]$$

$$I = (c+d)[\cosh(px)] + [(c-d)(\sinh(px))]$$

let,

$$a+b = A$$
$$a-b = B$$
$$c+d = C$$
$$c-d = D.$$

$$\boxed{V = A\cosh(px) + B\sinh(px)} \rightarrow (9)$$
$$\boxed{I = C\cosh(px) + D\sinh(px)} \rightarrow (10)$$

diff eqn (9) w.r.'x'. from eqn (1) $\frac{-dv}{dx} = IZ$

$$\frac{dv}{dx} = PA\sinh(px) + PB\cosh(px) \rightarrow$$

from eqn (1). $+\frac{dv}{dx} = -IZ$;

$$-IZ = P[A\sinh(px) + B\cosh(px)]$$

$$I = \frac{-P}{Z}[A\sinh(px) + B\cosh(px)]$$

WKT, $P = \sqrt{ZY}$, $Z_0 = \sqrt{\frac{Z}{Y}} \Rightarrow \frac{1}{Z_0} = \sqrt{\frac{Y}{Z}}$

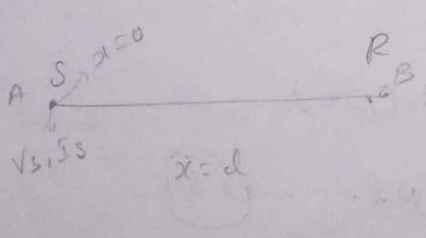
$$I = -\sqrt{\frac{Y}{Z}}[A\sinh(px) + B\cosh(px)]$$

$$\boxed{I = -\frac{1}{Z_0}[A\sinh(px) + B\cosh(px)]} \rightarrow (11)$$

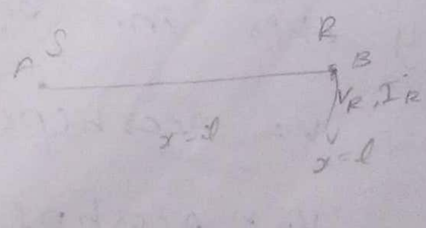
two case study to find A and B values.

if consider sending pt given i/p and vice versa

case (i)



case (ii)



case i) consider sending end. only known in sending value not recieving end, through that find A and B.

$x = 0, \quad V = V_s, \quad I = I_s.$

This conditions apply in eqn 9

9 =>

$$V = A\cosh(px) + B\sinh(px)$$

$$V_s = A$$

and lly apply in 11.

$$I_s = -\frac{B}{Z_0}$$

$$B = -I_s Z_0$$

apply A and B value in eqn 11 & 9

$$I = -\frac{1}{Z_0}\left[V_s \sinh(px) - I_s Z_0 \cosh px\right]$$

$$\boxed{V = V_s \cosh(px) - I_s Z_0 \sinh(px)} \longrightarrow (12)$$

$$\boxed{I = I_s \cosh px - \frac{V_s}{Z_0}\sinh(px)} \longrightarrow (13)$$

the ~~total~~ Total voltage and current is calculated with respect to sending.

Q: calculate total V and I of transmission line w.r.t sending.

consider receiving end (case ii)

$x = l, \quad V = V_R, \quad I = I_R$

apply this in eqn 9. $\quad \cosh^2\theta - \sinh^2\theta = 1$

9 =>

$$V = A\cosh(px) + B\sinh(px)$$

$$V_R = A\cosh pl + B\sinh pl. \longrightarrow (14)$$

in eqn (9)

$$I_R = -\frac{1}{Z_0}\left[A\sinh(pl) + B\cosh(pl)\right] \rightarrow (15)$$

$$(14) \times \frac{\cosh(pl)}{Z_0} \quad ; \quad (15) \times \sinh(pl)$$

(14) =>

$$V_R \cdot \frac{\cosh(pl)}{Z_0} = \frac{A\cosh^2(pl)}{Z_0} + \frac{B\sinh(pl)\cosh(pl)}{Z_0} \rightarrow (16)$$

(15) =>

$$\sinh(pl)\, I_R = -\frac{1}{Z_0} A\sinh^2(pl) - \frac{B\cosh(pl)\sinh(pl)}{Z_0} \rightarrow (17)$$

Add (16) + (17)

$$\frac{V_R\cosh(pl)}{Z_0} + \sinh(pl) I_R = \frac{A\cosh^2(pl)}{Z_0} - \frac{A\sinh^2 pl}{Z_0}$$

$\cosh^2 pl - \sinh^2 pl = 1$

$$= \frac{A}{Z_0}(1)$$

$$\boxed{A = V_R \cosh(pl) + I_R Z_0 \sinh(pl)}$$

sub A value in eqn (14)

$$(14) \times \frac{\sinh(pl)}{Z_0} \quad ; \quad (15) \times \cosh(pl)$$

$$(14) \Rightarrow \frac{V_R\sinh(pl)}{Z_0} = \frac{A\cosh(pl)\sinh(pl)}{Z_0} + \frac{B\sinh^2 pl}{Z_0} \rightarrow (18)$$

$$(15) \Rightarrow I_R\cosh(pl) = -\frac{A\sinh(pl)\cosh(pl)}{Z_0} - \frac{B\cosh^2 pl}{Z_0} \rightarrow (19)$$

Add eqn (18) & (19)

$$\frac{V_R\sinh(pl) + I_R\cosh(pl)}{Z_0} = \frac{B\sinh^2 pl}{Z_0} - \frac{B\cosh^2 pl}{Z_0}$$

$$\frac{V_R \sinh(pl)}{Z_0} + I_R \cosh(pl) = \frac{-B}{Z_0}$$

$$\boxed{B = -V_R \sinh(pl) - I_R Z_0 \cosh(pl)}$$

sub eqn A and B in 9 & 11

9 $\Rightarrow$

$$V = V_R \cosh(pl) + I_R Z_0 \sinh(pl) [\cosh(px)] +$$
$$- [V_R \sinh(pl) + I_R Z_0 \cosh(pl)] \sinh(px)$$

$\cosh(a+b) = \cosh a \cosh b + \sinh a \sinh b$

$$V = V_R \cosh^2(pl) + I_R Z_0 \cosh(pl) \sinh(pl)$$
$$- V_R \sinh^2(pl) - I_R Z_0 \cosh(pl) \sinh(pl)$$

$$V = V_R \cosh(pl) \cosh(px) + I_R Z_0 \sinh(pl) \cosh(px)$$
$$- V_R \sinh(pl) \sinh(px) - I_R Z_0 \cosh(pl) \sinh(px)$$

$$V = V_R [\cosh[pl - px]] + I_R Z_0 [\sinh(pl - px)]$$

$$V = V_R [\cosh p(l-x)] + I_R Z_0 [\sinh p(l-x)]. \rightarrow 20$$

lly,
$$I = I_R \cosh p(l-x) + \frac{V_R}{Z_0} \sinh p(l-x) \rightarrow 21$$

the total voltage and current is calculated with respect to receiving end.

Find sending end voltage and current.

conditions: $x=0$, $V=V_s$, $I=I_s$

sub these conditions in (20) & (21).

(20) $\Rightarrow$ $V = V_R\left[\cosh p(l-x)\right] + I_R Z_0\left[\sinh p(l-x)\right]$ } line eqn of voltage and current

$I =$ (21)

$$\boxed{\begin{aligned} V_s &= V_R \cosh pl + I_R Z_0 \sinh pl \\ I_s &= I_R \cosh pl + \frac{V_R}{Z_0}\sinh pl \end{aligned}}$$

sending voltage

sending current

Find receiving end voltage and current.

condition; $x=l$, $V=V_R$, $I=I_R$ → line eqn of V & I

sub these conditions in (12) & (13)

$$\boxed{\begin{aligned} (12) \Rightarrow \quad V_R &= V_s \cosh pl - I_s Z_0 \sinh pl \\ (13) \Rightarrow \quad I_R &= I_s \cosh pl - \frac{V_s}{Z_0}\sinh pl \end{aligned}}$$

Receiving voltage

Receiving current

Input impedance of a Transmission Line (Z_s or Z_{in})

sending impedance

$$Z_s = \frac{\text{sending voltage}}{\text{sending current}}$$

$$Z_s = \frac{V_s}{I_s}.$$

$$\boxed{Z_s = \left[\frac{V_R \cosh pl + I_R Z_0 \sinh pl}{I_R Z_0 \cosh pl + V_R \sinh pl}\right] . Z_0}$$

Transfer impedance

$$Z_s = \left[\frac{V_R\left[\frac{e^{pl}+e^{-pl}}{2}\right] + I_R Z_0\left[\frac{e^{pl}-e^{-pl}}{2}\right]}{I_R Z_0\left[\frac{e^{pl}+e^{-pl}}{2}\right] + V_R\left[\frac{e^{pl}-e^{-pl}}{2}\right]}\right] Z_0$$

$$Z_s = \left[\frac{V_R\left[e^{Pl}+e^{-Pl}\right] + I_R Z_0\left[e^{Pl}-e^{-Pl}\right]}{I_R Z_0\left[e^{Pl}+e^{-Pl}\right] + V_R\left[e^{Pl}-e^{-Pl}\right]}\right] Z_0$$

$$Z_s = Z_{in} = Z_0\left[\frac{e^{Pl}\left[V_R + I_R Z_0\right] + e^{-Pl}\left[V_R - I_R Z_0\right]}{e^{Pl}\left[I_R Z_0 + V_R\right] + e^{-Pl}\left[I_R Z_0 - V_R\right]}\right]$$

sub $V_R = I_R Z_R$. not perfectly matched.

If perfectly $V_R = I_R Z_{in}$

$Z_{in} = Z_0 = Z_R$

$$Z_{in} = Z_0\left[\frac{\left[I_R Z_R + I_R Z_0\right]e^{Pl} + e^{-Pl}\left[I_R Z_R - I_R Z_0\right]}{e^{Pl}\left[I_R Z_0 + I_R Z_R\right] + e^{-Pl}\left[I_R Z_0 - I_R Z_R\right]}\right]$$

$$= Z_0\left[\frac{I_R\left[Z_R + Z_0\right]e^{Pl} + I_R\left[Z_R - Z_0\right]e^{-Pl}}{I_R\left[Z_R + Z_0\right]e^{Pl} + I_R\left[Z_0 - Z_R\right]e^{-Pl}}\right]$$

$$\boxed{Z_{in} = Z_0\left[\frac{\left[Z_R + Z_0\right]e^{Pl} + \left[Z_R - Z_0\right]e^{-Pl}}{\left[Z_R + Z_0\right]e^{Pl} + \left[Z_0 - Z_R\right]e^{-Pl}}\right]}$$

Transfer impedance: (Z_T)

$$Z_T = \frac{\text{sending voltage}}{\text{Receiving current}} = \frac{V_s}{I_R}.$$

$$Z_T = \frac{V_R\cosh Pl + I_R}{}$$

from sending voltage eqn

$$V_s = V_R\cosh Pl + I_R Z_0 \sinh Pl.$$

since it is not perfectly matched condition,

then sub, $V_R = I_R Z_R$.

$$V_s = I_R Z_R \cosh pl + I_R Z_0 \sinh pl$$

$$V_s = I_R [Z_R \cosh pl + Z_0 \sinh pl]$$

$$\boxed{Z_T = \frac{V_s}{I_R} = Z_R \cosh pl + Z_0 \sinh pl.}$$

Reflection (K)

once when happened mismatching, then reflection signal is occur at the receiving end.

if perfectly matching, reflection is zero..

$$K = \frac{V_r}{V_i} = \frac{\text{reflected voltage}}{\text{incident voltage}}$$

$K = 1 \rightarrow$ open
$K = -1$ short

1) The reflection of energy occurs when there is an impedance irregularity that is when the primary constant of the transmission lines are not uniform along the line

2) The reflection takes place ~~based of~~ bcz the ratio of electric field and magnetic field on both the sides of terminals boundary are not the same under all conditions.

3) The conditions will be satisfied by the addition of reflected wave to the incident wave

4) The reflection coefficient ratio of amplitude of reflected and incident voltage waves at the receiving end of the line is frequently called

reflection coefficient.

$x=l$ $\quad -x=y$

$x=0$

$$k = \frac{V_r}{V_i}$$

(or)

$$K = -\frac{I_r}{I_i}$$

$V = ae^{px} + be^{-px}$

$y = -x$

$x = -y$

$V = ae^{-py} + be^{py}$

$I = ce^{px} + de^{-px}$

$y = -x \; ; \; x = -y$

$I = ce^{-py} + de^{py}$

At the receiving end point

This is equal to

apply $x=-y$

$$I = -\frac{1}{Z_0}\left[ae^{px} - be^{-px}\right] \Rightarrow -\frac{1}{Z_0}\left[ae^{-py} - be^{py}\right]$$

WKT,

$$I = -\frac{1}{Z_0}\left[\underset{a+b}{A}\sinh(px) + \underset{a-b}{B}\cosh(px)\right]$$

$$= \frac{-1}{Z_0}\left[a\sinh px + b\sinh px + a\cosh px - b\cosh px\right]$$

$$= \frac{-1}{Z_0}\left[a\left[\sinh px + \cosh px\right] + b\left[\sinh px - \cosh px\right]\right]$$

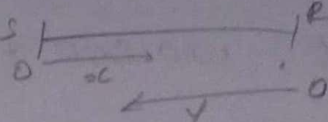
$$\boxed{I = -\frac{1}{Z_0}\left[ae^{px} - be^{-px}\right]}$$

Therefore,

$$V = ae^{-py} + be^{py} \longrightarrow (1)$$

$$I = -\frac{1}{Z_0}\left[ae^{-py} - be^{py}\right] \longrightarrow (2)$$

At the receiving end, $y=0$,



$V = V_R$, $I = I_R$.

These are apply on (1) & (2)

$V_R = ae^{-p(0)} + be^{p(0)}$

$V_R = a + b \longrightarrow (3)$

$I_R = -\frac{1}{Z_0}\left[ae^{0} - be^{0}\right]$

$= -\frac{1}{Z_0}\left[a - b\right]$

$I_R = \frac{b}{Z_0} - \frac{a}{Z_0} \longrightarrow (4)$

(4) $\Rightarrow$

$I_R Z_0 = b - a \longrightarrow (5)$

solve eqn (3) & (5)

(3) + (5)

$V_R = \cancel{a} + b$

$I_R Z_0 = b - \cancel{a}$

$V_R + I_R Z_0 = 2b$

$$\boxed{b = \frac{V_R + I_R Z_0}{2}}$$

(3) (5)

$V_R = a + \cancel{b}$

$-I_R Z_0 = -\cancel{b} + a$

$V_R - I_R Z_0 = 2a$

$$\boxed{a = \frac{V_R - I_R Z_0}{2}}$$

$k = \frac{V_r}{V_i}$

from eqn (1), $ae^{-py} = V_r$, $V_i = be^{py}$

$k = \frac{ae^{-py}}{be^{py}} = \frac{a}{b}e^{-2py}$; at receiving end $y=0$

$k = \frac{a}{b}$

$$\boxed{k = \frac{V_R - I_R Z_0}{V_R + I_R Z_0}}$$

Though, we see the conditions for not perfectly matched, therefore

$$V_R = I_R Z_R$$

$$K = \frac{I_R Z_R - I_R Z_0}{I_R Z_R - I_R Z_0}$$

$$= \frac{\cancel{I_R}\left[Z_R - Z_0\right]}{\cancel{I_R}\left[Z_R + Z_0\right]}$$

$$\boxed{K = \frac{Z_R - Z_0}{Z_R + Z_0}}$$

WKT, eqn for i/p impedance

if K is given & asked to find i/p impedance

$$Z_{in} = Z_0 \left[\frac{(Z_R + Z_0)e^{Pl} + (Z_R - Z_0)e^{-Pl}}{(Z_R + Z_0)e^{Pl} - (Z_R - Z_0)e^{-Pl}}\right]$$

$$= Z_0\left(\frac{Z_R + Z_0}{Z_R + Z_0}\right)\left[\frac{e^{Pl} + \left(\frac{Z_R - Z_0}{Z_R + Z_0}\right)e^{-Pl}}{e^{Pl} - \left(\frac{Z_R - Z_0}{Z_R + Z_0}\right)e^{-Pl}}\right]$$

$$= Z_0\left[\frac{e^{Pl} + Ke^{-Pl}}{e^{Pl} - Ke^{-Pl}}\right] \quad ; K = \frac{Z_R - Z_0}{Z_R + Z_0}$$

$$Z_{in} = Z_0\left[\frac{1 + Ke^{-2Pl}}{1 - Ke^{-2Pl}}\right]$$

where $K \rightarrow$ reflection coefficient

$P \rightarrow$ propagation constant

$Z_0 \rightarrow$ trans impedance

The above eqn I/p impedance can be expressed in term of reflection coefficient.

open circuit and short circuit line

w.k.t, input impedance eqn,

$$Z_{in} = Z_0 \left[\frac{(Z_R + Z_0)e^{Pl} + (Z_R - Z_0)e^{-Pl}}{(Z_R + Z_0)e^{Pl} - (Z_R - Z_0)e^{-Pl}} \right]$$

$$Z_s = Z_0 \left[\frac{V_R \cosh Pl + I_R Z_0 \sinh Pl}{I_R Z_0 \cosh Pl + V_R \sinh Pl} \right]$$

$$Z_{in} = Z_0 \left[\frac{Z_R \cosh(Pl) + Z_0 \sinh(Pl)}{Z_0 \cosh(Pl) + Z_R \sinh(Pl)} \right]$$

OC ;

$Z_R = \infty$

$$Z_{in} = \frac{Z_0 Z_R}{Z_R} \left[\frac{\cosh Pl + \frac{Z_0}{Z_R} \sinh Pl}{\frac{Z_0}{Z_R} \cosh Pl + \sinh Pl} \right]$$

$Z_R = \infty$

$$= Z_0 \left[\frac{\cosh Pl + 0}{0 + \sinh Pl} \right]$$

$$Z_{oc} = \left(\frac{\cosh Pl}{\sinh Pl} \right) Z_0.$$

$$\boxed{Z_{oc} = Z_0 \coth(Pl)}$$

SC ; $Z_R = 0$

$$Z_{in} = Z_0 \left[\frac{Z_0 \sinh Pl}{Z_0 \cosh Pl} \right]$$

$$\boxed{Z_{sc} = Z_0 \tanh(Pl)}$$

$Z_{oc} \times Z_{sc}$

$$Z_{oc} \times Z_{sc} = Z_0^2 \coth(Pl) \times \tanh Pl)$$

$$= Z_0^2 \left[\frac{\cosh Pl}{\sinh Pl} \right] \times \left[\frac{\sinh Pl}{\cosh Pl} \right]$$

$Z_{oc} Z_{sc} = Z_0^2$

$$\boxed{Z_0 = \sqrt{Z_{oc} \cdot Z_{sc}}}$$

Z_0 in terms of open & short ckt impedance.

~~wkt,~~ $Z_0 = \sqrt{Z/Y}$.

The above Eqn represents characteristic impedance can be expressed in terms of Z_{sc} & Z_{oc}.

24/6/26

1. The characteristics impedance an uniform transmission line is 2039.5 Ω at frequency & 800HZ. At this frequency the propagation constant was found to be $0.054 \angle 87.9°$. Determine the value of line constants.

Soln. Line constants → R, L, C, G.

Given;

$Z_0 = 2039.5\ \Omega$

$P = 0.054 \angle 87.9°$

$f = 800\ HZ$

$\omega = 2\pi f$

wkt,

$Z = R + j\omega L$

$Y = G + j\omega C$

$P = \sqrt{ZY}$; $Z_0 = \sqrt{\frac{Z}{Y}}$

$P \times Z_0 = \sqrt{Z}\sqrt{Y} \times \frac{\sqrt{Z}}{\sqrt{Y}}$

$P Z_0 = Z$

$$\frac{P}{Z_0} = \frac{\sqrt{ZY}}{\sqrt{\frac{Z}{Y}}}$$

$$= \frac{\sqrt{Z}\sqrt{Y}\cdot\sqrt{Y}}{\sqrt{Z}}$$

$$\frac{P}{Z_0} = Y.$$

$(a\angle b)(c\angle d)$

$= (a\times c)\angle(b+d)$

$\frac{a\angle b}{c\angle d} = (a\div c)\angle b-d$

$$PZ_0 = Z = (0.054\angle 87.9)(2039.5)$$

$$= (1.9787\times10^{-3} + j0.054)(2039.5)$$

(or) $110.133\angle 87.9$ $Z = 4.035 + j110.06$

$$Y = \frac{P}{Z_0} = \frac{0.054\angle 87.9^\circ}{2039.5\angle 0^\circ}$$

$$= \cancel{110.133}$$

$$= 2.648\times10^{-5}\angle 87.9$$

$$Y = 9.70\times10^{-7} + 2.65\times10^{-5}j$$

$$Z = R + j\omega L = 4.035 + 110.06j$$

$$Y = G + j\omega C = 9.7\times10^{-7} + 2.65\times10^{-5}j$$

From eqn; $\boxed{R = 4.035\ \Omega/km}$ $\boxed{G = 9.7\times10^{-7}\ \mho/km}$

$\omega L = 110.06$ | $\omega C = 2.65\times10^{-5}$

$(2\pi f)L = 110.06$ | $(2\pi f)\ C = 2.65\times10^{-5}$

$L = \frac{110.06}{2\pi(800)}$ | $C = \frac{2.65\times10^{-5}}{2\pi(800)}$

$\boxed{L = 0.0219\ H/km}$ $\boxed{C = 5.272\times10^{-9}\ F/km}$

2. The characteristic impedance of an certain line is $710\angle -16°$ when the frequency is 1KHZ, at this frequency the attenuation is 0.01 NP/km and phase constant is 0.035 rads/km. Calculate the R, L, C and G.

Soln:- Given:

$Z_0 = 710\angle -16°$

$\alpha = 0.01$ Neper/km

$\beta = 0.035$ radians/km.

$P = \alpha + j\beta$

$P = 0.01 + j0.035$

$P = 0.036\angle 74.054$.

$Z = PZ_0 = (0.036\angle 74.054)(710\angle -16°)$

$= 25.56\angle \cancel{90.054}$ 58.054.

$\cancel{Z = -0.024 + 25.56j}$

$Z = 13.524 + 21.69j$

$Y = \frac{P}{Z_0} = \frac{0.036\angle 74.054}{(710\angle -16)}$

$= 5.07\times 10^{-5}\angle 90.054$

$Y = -4.79\times 10^{-8} + 5.07\times 10^{-5}$

$Z = R + j\omega L = 13.524 + 21.69j$

$\boxed{R = 13.524}$ Ω/km $\quad \omega L = 21.69$

$2\pi f L = 21.69$

$L = \frac{21.69}{2\pi(1K)} = 3.452\times 10^{-3}$ H/km

$y = G + j\omega C = -4.79\times10^{-8} + 5.07\times10^{-5}$

$$\boxed{G = -4.79\times10^{-8}\ \mho/km}$$

$\omega C = 5.07\times10^{-5}$

$$C = \frac{5.07\times10^{-5}}{2\pi\times1000}$$

$$\boxed{C = 8.069\times10^{-9}\ F/km.}$$

3. openckt and short ckt impedance of an transmission line at 1.6KHZ are $900\angle-30^\circ$ and $400\angle-10^\circ$ calculate Z_0.

soln.

given,

$Z_{OC} = 900\angle-30^\circ$, $Z_{SC} = 400\angle-10^\circ$

$f = 1.6KHZ$. , $Z_0 = ?$

$$Z_0 = \sqrt{Z_{OC} Z_{SC}}$$

$$= \sqrt{(900\angle-30^\circ)(400\angle-10^\circ)}$$

$$Z_0 = 600\angle-20^\circ = 563.82 - 205.21j$$

4. The 100 km long transmission line is terminated by 100 Ω. It has the following constants the resistance 200 Ω. It has the following constants $Z_0 = 600\,\Omega$, $\alpha = 0.01$ Neper/km, $\beta = 0.03$ rad/km. Find the input impedance. and reflection coefficient.

Soln

$P = \alpha + j\beta$

$P = 0.01 + j\,0.03$

$Z_0 = 600\,\Omega$

$l = 100$

$Z_R = 200$

$$K = \frac{Z_R - Z_0}{Z_R + Z_0} = \frac{200-600}{200+600} = \frac{-400}{800} = \frac{-1}{2}$$

$= 0.5\angle180$

- indicates opp direction

$$Z_{in} = Z_0 \left[\frac{1 + |K| e^{-2Pl}}{1 - |K| e^{-2Pl}} \right] \qquad \frac{1 - 0.5\, e^{-200(0.5\angle 180)}}{1 + 0.5\, e^{-200(0.5\angle 180)}}$$

$$Z_{in} = 600 \left[\frac{1 + (-0.5) e^{-2(0.01 + j0.03)100}}{1 - (-0.5) e^{-2(0.01 + j0.03)100}} \right]$$

$$Z_{in} = 600 \left[\frac{1 - 0.5\, e^{-2-6j}}{1 + 0.5\, e^{-2-6j}} \right]$$

$$\boxed{|K| = 0.5 \angle 180^\circ}$$

mag = $\sqrt{real^2 + img^2}$

$e^{Pl} =$

$\boxed{1\, rad = 57.3}$

$$e^{-2Pl} = e^{-2-6j}$$

$$= e^{-2} \cdot e^{-6j}$$

$$= 0.1353 \cdot [\cos(-6) - j\sin(-6)]$$

convert to rad. bcz (β is rad)

$$= 0.1353\, [0.9945 - j(-0.1045)]$$

$$= 0.1353\, [0.960 - j(-0.279)]$$

$$e^{-2Pl} = 0.1353\, [0.960 + j0.279]$$

$$= 0.1298 + j0.0377$$

(or)

$6 \times 57.3 = 343.8$ (in degree)

$$e^{-2Pl} = 0.1353\, [\cos(343.8) - j\sin(343.8)]$$

$$= 0.1353\, [0.9602 - j(-0.2789)]$$

$$= 0.1353\, [0.9602 + j0.2789]$$

$$e^{-2Pl} = 0.1298 + j0.0377$$

$$Z_{in} = 600\left[\frac{1+0.5\,(0.1298+j0.0377)}{1-0.5\,(0.1298+j0.0377)}\right]$$

$$= 600\left[\frac{1+[0.0649+j0.01885]}{1-[0.0649-j0.01885]}\right] \qquad = \frac{1.0649+j0.01885}{0.935\,1\,j\,0.01885}$$

$$= 600\left[\frac{\cancel{0.9351+j0.01885}}{\cancel{0.9351+j0.01885}}\right] \qquad \frac{1.0649\angle 0.56}{0.9352\angle -1.155}$$

$$= 600\left[\frac{0.935\angle 1.155}{1.065\angle 1.022}\right] \qquad 2.0692\angle 1.715$$

$$= 600\,[0.88\angle 0.133] \qquad 1241.52\angle 0.133$$

$$= [600\angle 0][0.88\angle 0.133]$$

$$Z_{in} = 528\angle 0.133 = 527.9 + j1.225$$

$Z_{in} = 682.61 + 25.783]$

$(1.1379 + j0.0430)\,600$

$\boxed{682.74 + j\,25.8}$

25/6/26

5. A transmission line of 50km has a characteristic impedence $692\angle{-12°}$, phase constant 0.0355 rad/Km, attenuation const, 0.00755 Np/Km, sending end voltage is 10V at freq of 1KHZ and the line is terminated by a 300Ω resistor. calculate sending end current, receiving end current and voltage.

Soln Given:

$l = 50\,km$ | $f = 1\,KHZ$
$\beta = 0.0355$ | $Z_R = 300$
$Z_0 = 692\angle -12°$ | $I_S = ?$
$\alpha = 0.00755$ | $V_R = ?$
$V_S = 10\,V$ | $I_R = ?$

$P = \alpha + j\beta$

$P = 0.00755 + j0.0355$

$P = 0.0361\angle 77.99$

$$Z_{in} = Z_0 \left[\frac{[Z_R + Z_0] e^{Pl} + [Z_R - Z_0] e^{-Pl}}{[Z_R + Z_0] e^{Pl} + [Z_0 - Z_R] e^{-Pl}} \right]$$

$\boxed{Z_{in} = Z_S}$

$$e^{Pl} = e^{(0.0361 \angle j77.99) 50}$$

$$= e^{1.805 \angle 127.99j}$$

$$= e^{\cancel{1.805}} e^{(-1.11 + j1.4225)}$$

$$= e^{-1.11} \cdot e^{j1.4225}$$

$$= e^{-1.11} \cdot [\cos(1.4225) + j\sin(1.4225)]$$

$$= 0.31 [0.99 + j0.025]$$

$$= \cancel{0.31 [1.025]} = 0.3069 + j\,7.75 \times 10^{-3}$$

$$e^{Pl} = \cancel{0.31775} \qquad = 0.307 \angle 1.446.$$

$$e^{-Pl} = -0.31775$$

$$Z_{in} = 692 \angle -12 \left[[300 + (692 \angle -12)] \right]$$

$$Z_{in} = Z_0 \left[\frac{Z_R \cosh Pl + Z_0 \sinh Pl}{Z_0 \cosh Pl + Z_R \sinh Pl} \right]$$

$$Pl = (0.0361 \angle j\,77.9)(50 \angle 0)$$

$$= (1.815 \angle 77.9) \longrightarrow 0.3775 + j1.775$$

$(1.8 \cdot 34 \angle 77.9)$

$$\cosh pl = \frac{e^{pl} + e^{-pl}}{2} \; ; \; \sinh pl = \frac{e^{pl} - e^{-pl}}{2}$$

$$e^{pl} = e^{(0.035 \angle T)}$$

$$e^{pl} = 0.3775 + j1.775$$

$$\cosh pl = \cosh(0.3775 + j1.775)$$

$$\cos(a+b)$$

$$= \left(\cosh(0.3775)\cosh(j1.775)\right) + \left(\sinh(0.3775)\sinh(j1.775)\right)$$

$$\because \cosh(j\theta) = \cos\theta$$

$$\sinh(j\theta) = j\sin\theta$$

$$\cosh pl = \cosh(0.3775)\cos(1.775) + \sinh(0.3775)\, \sin j(1.775)$$

rad

$$= (1.072)(-0.20) + (0.39)(j0.979)$$

$$\cosh pl = -0.2144 + 0.3818j = 0.44 \angle 119.3$$

$$\sinh pl = \sinh(0.3775 + j1.775)$$

$$= \sinh(0.3775)\cos(j1.775) + \cosh(0.3775)\, j\sin(j1.775)$$

$$= 0.39(j-0.20) + 1.072(j0.979)$$

$$\sinh pl = -0.078 + j1.0495$$

$$= 1.052 \angle 94.25$$

$$Z_{in} = 692\angle -12 \left[\frac{300[0.44\angle 119.3] + (692\angle -12)[1.052\angle 94.25]}{692\angle -12[0.44\angle 119.3] + 300[1.052\angle 94.25]} \right]$$

$$= 692\angle -12 \left[\frac{132\angle 119.3 + 727.984\angle 82.25}{304.48\angle 107.3 + 315.6\angle 394.25} \right]$$

$$= 692\angle -12 \left[\frac{67.392 + j113.5 + 98.2 + j721.33}{-90.54 + j290.7 + 260.9 + j177.62} \right]$$

$$= 692\angle -12 \left[\frac{165.59 + j834.83}{170.36 + j468.32} \right]$$

$$= 692\angle -12 \left[\frac{851.09\angle 78.78}{498.34\angle 70.0} \right]$$

$$= 692\angle -12 \left[\overset{1.3242 - j0.3046}{\cancel{1.707\angle 8.78}} \right]$$

$$= \overset{852.5 - j396.69}{\cancel{1181.244\angle -3.22}}$$

$$Z_{in} = \cancel{1179.38 - 66.3j} \quad 940.27\angle -25$$

~~$I_s = \frac{V_s}{Z_{in}} = \frac{10}{989.5\angle -25}$~~

$Z_0 \cosh \gamma l = (692\angle -12)(-0.2144 + j0.3822)$

$= -90.24 + j290.70$

$Z_R \sinh \gamma l = (300)(-0.078 + j1.05)$

$= -23.4 + j315$

$Z_R \cosh \gamma l = (300)(-0.2144 + j0.3822)$

$= -64.32 + j114.66$

$Z_0 \sinh \gamma l = (692\angle -12)(-0.078 + j1.05)$

$= 98.272 + j721.94$

$Z_{in} = (692\angle -12)\left[\frac{-64.32 + j114.66 + 98.272 + j721.94}{-90.54 + j290.70 - 23.4 + j315}\right]$

$Z_{in} = (692\angle -12)\left[\frac{33.952 + j836.6}{-113.94 + j605.7}\right]$

$Z_{in} = (692\angle -12)\left[1.323 - j0.305\right]$

$Z_{in} = 851.63 - j396.7$

$Z_{in} = 939.5\angle -25$

$I_s = \frac{V_s}{Z_{in}} = \frac{10}{939.5\angle -25} = 0.0106\angle 25 = 9.606\times 10^{-3} + j4.479\times 10$

$V_R = V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l$

$= 10[0.44\angle 119.3] - [0.0106\angle 25\,(98.272 + j721.94)]$

$V_R = 0.1367 - j3.538$

$V_R = 3.541\angle -87.79$

$I_R = \frac{V_R}{Z_R} = \frac{0.1367 - j3.538}{300}$

$I_R = 4.556 \times 10^{-4} - j0.012$

$I_R = 0.012 \angle -87.826$

-88

29/6/26

assignment

1. a Transmission line 50km long gave the following results of input impedance at 796 Hz, open ckt impedance $328 \angle -29.2^\circ$ and short ckt impedance $1548 \angle 6.8$ determine Line constant (R, L, C, G)

Soln given: $l = 50 km$

$Z_{oc} = 328 \angle -29.2$

$Z_{sc} = 1548 \angle 6.8$

$f = 796 Hz.$

$Z_0 = \sqrt{Z_{oc} \cdot Z_{sc}}$

$= \sqrt{(328 \angle -29.2)(15.98 \angle 6.8)}$

$Z_0 = 712.562 \angle -11.2$

$Z_0 = 698.9 - j138.40$

$\frac{Z_{sc}}{Z_{oc}} = \frac{Z_0 \tanh(Pl)}{Z_0 \coth(Pl)}$

$= \frac{\cancel{Z_0} \tanh(Pl)}{\cancel{Z_0} \frac{1}{\tanh}(Pl)}$

$\frac{Z_{sc}}{Z_{oc}} = \tanh^2(Pl)$

$\frac{1548 \angle 6.8}{328 \angle -29.2} = \tanh^2(Pl)$

$\sqrt{4.7195 \angle 36} = \tan^{*}h(pl)$ $2.17 \angle 18$

Rec $2.06 + j0.671 = P$

$\tanh(pl) = 2.17 \angle 18$

$$\frac{\sinh pl}{\cosh pl} = \frac{e^{pl} - e^{-pl}/\cancel{2}}{e^{pl} + e^{-pl}/\cancel{2}} = 2.17 \angle 18$$

$$= \frac{e^{pl} - e^{-pl}}{e^{pl} + e^{-pl}} = 2.17 \angle 18$$

$$\frac{N}{D} = \frac{D+N}{D-N}$$ componend dividend method

$$\frac{e^{pl} + e^{-pl} + e^{pl} - e^{-pl}}{e^{pl} + e^{-pl} - e^{pl} + e^{-pl}} = \frac{1 + 2.06 + j0.671}{1 - 2.06 - j0.671}$$

$$\frac{2e^{pl}}{2e^{-pl}} = \frac{3.06 + j0.671}{-1.06 - j0.671}$$

$$e^{2pl} = -2.35 + j0.853$$

$$e^{2pl} = 2.5 \angle 160$$

Take both side natural ln.

$$2pl = \ln(2.5 \angle 160)$$

angle $+ j2\pi n$

1 rad → deg $\times 57.3$

deg → rad $\div 57.3$

$$2pl = 0.916 \angle 166.28\,n$$

$$P = \frac{0.916 \angle 166.28\,n}{100}$$

$$P = 9.16 \times 10^{-3} \angle 166.28\,n$$

$2Pl = \ln(2.5 \angle 160)$

$2Pl = 0.916 + j(160 + 2\pi n)$

$P = 9.16\times10^{-3} + j(2.79 + 2(\cancel{3.14})n)\times10^{-2}$

$P = 9.16\times10^{-3} + j(2.79 + 6.28n)\times10^{-2}$

β value should be +ve and lowest value

when apply values for n.

Hence choose $n=0$,

$P = 9.16\times10^{-3} + j(2.79)\times10^{-2}$

$0.00916 + 0.0279$

→ when $n=0$ β values gives 2.79×10^{-2}

→ when $n=0$ β value is both positive and lowest value.

From eqn of P

$P \cdot Z_0 = Z$

$Z = (9.16\times10^{-3} + j2.79\times10^{-2})(698.9 - j138.40)$

$Z = 10.264 + j18.234$

$\frac{P}{Z_0} = Y$

$$Y = \frac{(9.16\times10^{-3} + j2.79\times10^{-2})}{(698.9 - j138.40)}$$

$Y = 5.0\times10^{-6} + j4.1\times10^{-5}$

$Z = R + j\omega L$

$10.264 + j18.234 = R + j\omega L$

$\boxed{R = 10.264}$ Ω/km

$\omega L = 18.234$

$L = \frac{18.234}{2\pi(796)}$

$L = 3.645\times10^{-3}$ H/km

$y = G + j\omega C$

$5.0\times10^{-6} + j\,4.1\times10^{-5} = G + j\omega C$

$G = 5.0\times10^{-6}$ $\qquad \omega C = 4.1\times10^{-5}$

$C = \dfrac{4.1\times10^{-5}}{2\pi(796)}$

$C = 8.194\times10^{-9}$

8×10^{-9}

2. The characteristic impedance of a certain line is $710\angle-16°$, when the frequency is 1000HZ. At this frequency attenuation function is 0.01 NP/km and phase function is 0.035 rad/km if the line 100 km long is terminated by a 300Ω resistor. calculate the ratio of sending voltage w.r.t respect to receiving voltage. (V_S/V_R)

Soln

Given:

$Z_0 = 710\angle-16°$, $\alpha = 0.01$ NP/km

$f = 1KHZ$, $\beta = 0.035$ rad/km

$l = 100$ km, $Z_R = 300$.

$P = 0.01 + j0.035$

$V_S = V_R \cosh Pl + Z_0 I_R \sinh Pl$.

$V_S = V_R\left[\cosh Pl + \dfrac{I_R Z_0 \sinh Pl}{V_R}\right]$

$\dfrac{V_S}{V_R} = \cosh Pl + \dfrac{Z_0}{Z_R}\sinh Pl.$

$\frac{V_S}{V_R}$ $pl = (0.01 + j0.035)(100)$

$pl = 1 + j3.5$

p.

$\cosh pl = \cosh(1 + j3.5)$

$= \cosh(1)\cosh(j3.5) + \sinh(1)\sinh(j3.5)$

$= \cosh(1)\cos(3.5) + \sinh(1)\, j\sin(3.5)$

$= (1.543)(-0.936) + 1.175(-j0.351)$

$\cosh pl = -1.444 - j0.4124$

$= 1.5017 \angle -164.06$

$\sinh pl = \sinh(1 + j3.5)$

$= \sinh(1)\cosh(j3.5) + \cosh(1)\sinh(j3.5)$

$= 1.175(-0.936) + 1.543(-j0.351)$

$\sinh pl = -1.0998 - j0.542 = 1.226 \angle -153.76$

$\frac{V_S}{V_R} = (-1.444 - j0.4124) + 2.275 + j0.652(-1.0998 - j0.542)$

$\downarrow$

$2.4\angle$

$\frac{Z_C}{Z_R} = \frac{710\angle -16}{300} = 2.274 \angle -0.65$

$\frac{V_S}{V_R} = -3.588 - j2.3625$

$\frac{V_S}{V_R} = 4.295 \angle -146.63$

$\frac{V_S}{V_R}$

$\frac{V_S}{V_R} = \cancel{1.5017 \angle -163.06}\ -4.5 - j1.616$

$\frac{V_S}{V_R} = 4.78 \angle -160.2$

Loading:

Def: Adding Inductance in series with Transmission line to reduce attenuation Loss is called loading.

	$\alpha \downarrow$	$Z_0 \uparrow$	
$R\downarrow$	$\alpha \uparrow$	$Z_0 \downarrow$	cost↑
$L\uparrow$	$\alpha \downarrow$	$Z_0 \uparrow$	cost↑
$G\uparrow$	$\alpha \downarrow$	$Z_0 \uparrow$	cost↑
$C\downarrow$	$\alpha \downarrow$	$Z_0 \uparrow$	cost↑

$RC = LG$

condition for distortion less line ← $\boxed{\frac{R}{L} = \frac{G}{C}}$ ideal case

signal is distorted ← $\boxed{\frac{R}{L} >> \frac{G}{C}}$ → practical case

Types:

continuous loading

TL
copper conductor
TL

Iron or some other magnetic material is wounded around the conductor to the loaded thus increasing the permeability of the surrounding medium.

$$L = \frac{\mu}{\frac{d}{nt} + 1} \text{ mH.}$$

μ → permeability of surrounding ~~medium~~ material

d → diameter of the copper material

n → no. of layers

t → thickness per layer

Advantages.

→ Attenuation is independent of frequency

$$\alpha = \frac{R}{2}\sqrt{\frac{C}{L}}$$

→ Attenuation is decrease by increasing inductance. (R is not increased)

→ Increasing in inductance upto 100 mH

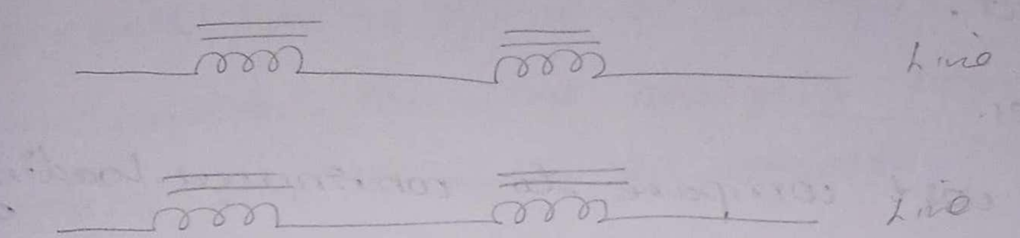
disadvantages

→ very expensive

→ existing line cannot modify by this method

only replacement of line.

Lumped method loading → only for a limited range of freq.



Def: The inductance of transmission line can be increased by insert a lumped inductance (toss) at a uniform intervals along the line

Lumped line act as LPF [so it will work in low frequency after when f↑ (internal) then happens eddy current loss & hysteresis loss

Adv

→ cost is low

→ existing line can be modified

disadvantages.

→ at certain freq, attenuation loss is ↑ then the $\boxed{\text{internal connected resistor value is } \uparrow}$, then create the eddy current loss and hysteresis loss.

By reduce this disadvantages [campbell's eqn)

$$\cosh N\gamma' = \cosh(N\gamma) + \frac{Z_L}{2Z_0}\sinh(N\gamma)$$

Patch loading

It is a method of increasing the inductance of a transmission line by loading only certain section of the cable continuously while leaving the remaining section unloaded.

Advantages:

→ lower cost compare to continuous loading

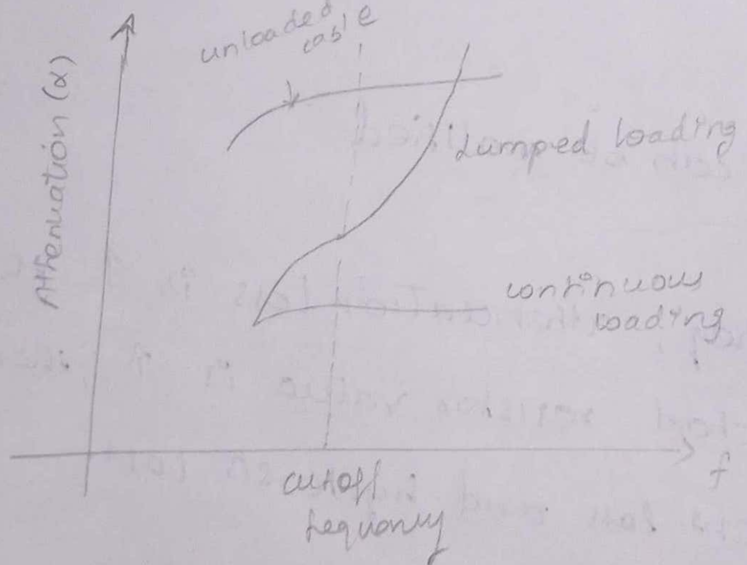
→ reduce attenuation loss

→ ~~include~~ improve long distance communication.

disadvantages:

→ performance is slightly less uniform than full continuous loading

attenuation freq characteristics graph



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Infinite line

→ The propagation of electric waves along uniform line is getting deduced due to the uncertain length of infinite line.

→ A signal fed into a line could not reach the far end at a finite line. the condition at the far end has no effect at a input end.

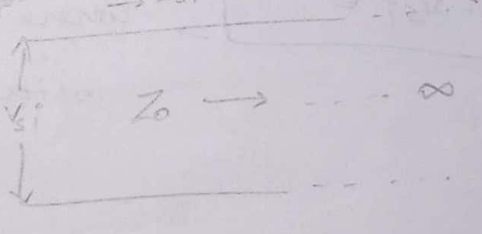
→ Transmission line analysis begins with infinite line in order to seperate input condition from output condition.

→ when AC voltage is applied to the sending end of an infinite line, a finite current will flow. capacitance and linkage conductance between the two wires of the line

→ The ratio of the voltage applied to the current flowing will give the i/p impedance of a infinite line

$$Z_0 = \frac{V_{si}}{I_{si}}$$

→ sending end voltage for a infinite line



why Z_0 is imp?
↓
at any particular point impedance is matching so we put Z_0 instead Z_0

current at any point distance x from the sending end why? bcz Zn receiving is infinite

$$I = ce^{px} + de^{-px} \rightarrow (1)$$

→ The value of small c and d can now determine by considering an infinite line.

at the sending end of the infinite line the condition is,

$x=0,\ I=I_{si}$

$I_{si} = c+d \quad \rightarrow (2)$

at receiving end, $x=\infty, I=0$ sub in eqn (1)

$0 = ce^{\infty} + de^{-\infty}$

$0 = ce^{\infty} + 0$

$0 = c(\infty)$

$c = \frac{0}{8}$, $\boxed{c=0}$

sub $c=0$ in (2)

$\boxed{I_{si} = d}$

sub value of c and d in (1)

$I = 0 + I_{si}e^{-px}$

$\boxed{I = I_{si}e^{-px}}$ where I_{si} ↓ sending end current for a infinite line

lly, $\boxed{V = V_{si}e^{-px}}$ → where V_{si} is sending end voltage for a infinite line.

Problem

1. The Transmission line of characteristic impedance $500\angle-43^\circ$ and propagation constant $0.07+j0.08$ per km is properly terminated. A voltage of $5\angle0^\circ$ volt is applied at sending end. calculate voltage and current at a distance of 10 km from the sending end.

$Z_0 = Z_{in} = Z_R = Z_s$

$\underline{\text{soln}}$ Given:

$Z_o = 500\angle{-43}^\circ$ $\quad V_{si} = 5\angle 0$

$P = 0.07 + j0.08$ $\quad \cancel{l} = 10\,km = x$.

$V = V_s\cosh Pl - I_s Z_o \sinh$

Properply termiated

$Z_o = Z_s = Z_R$

$V_s = I_s Z_o$

$\boxed{V_s = I_s Z_o}$

$\boxed{V = V_s[\cosh Pl - \sinh Pl]}$

$$Z_o = \frac{V_{si}}{I_{si}} \quad ; \quad I_{si} = \frac{V_{si}}{Z_o}$$

$$I_{si} = \frac{5\angle 0}{500\angle{-43}}$$

$$I_{si} = 0.01\angle 43.$$

$$V = V_{si}\, e^{-Px}$$

$$= (5\angle 0)\, e^{-(0.07+j0.08)10}$$

$$= (5\angle 0)\, e^{-0.7 - j0.8}$$

$$= (5\angle 0)\, e^{-0.7} \cdot e^{-j0.8}$$

$$= (5\angle 0)(0.4966)\left[\cos[0.8\times 57.3] - j\sin[0.8\times 57.3]\right]$$

$$= (2.483\angle 0)[0.7 - j0.717]$$

$$= (2.483\angle 0)(1.0\angle{-45.69})$$

$$\boxed{V = 2.483\angle{-45.69} = 1.734 - j1.778}$$

$V = IZ_o \quad I = \frac{V}{Z_o}$

$I = \frac{2.483\angle{-45.69}}{5\angle 0}$

$I = 0.49\angle{-45.69}$

$I =$

$$I = I_{si}\, e^{-Px}$$

$$= (0.01\angle 43)\, e^{-0.7 - j0.8}$$

$$= (0.01\angle 43)\, e^{-0.7} \cdot e^{-j0.8}$$

$$= (0.01\angle 43)(0.4966)[0.7 - j0.717]$$

$$= (4.96\times 10^{-3}\angle 43)(0.7 - j0.717)$$

$$\boxed{I = 4.964\times 10^{-3} - j2.33\times 10^{-4}.}$$

$4.965\times 10^{-3}\angle{-2.689}$

6/7/26.

Wavelength

The distance travelled by the wave along the line while the phase angle is changing through 2π radians is called wavelength. ①

$$\lambda = \frac{2\pi}{\beta} \quad \text{(or)} \quad \lambda = \frac{v}{f}$$

Velocity of propagation

It is defined as the velocity at which the signal of single frequency propagates along the line at a particular frequency.

$$v = \frac{\omega}{\beta} \quad / \quad v = \lambda f.$$ ②

Distortion:

→ A signal transmitted over a transmission line is normaly complex and consist of many frequency components.

→ such voice voltage will not have all frequencies transmitted with equal attenuation & equal time delay, the received waveform will not be identical with the input waveform at the sending end. This variation is known as distortion.

Types of Distortion

(1) → A complex voltage transmitted on a transmission line will not be attenuated equally and the receive waveform will not be identical with the input waveform at the transmitting end. This variation is known as frequency distortion.

→ α is a function of frequency, Therefore the line will introduced, frequency distortion.

dependent of frequency

→ The frequency distortion is reduced in the transmission of high quality over wire lines by the use of equalizers at the line terminations

(2) Delay or phase distortion

→ For an applied voice voltage wave the received waveform may not be identical with the input waveform at the sending end, since some frequency components will be delay more than those of other frequencies. This phenomenon is known as phase or delay distortion

→ Delay distortion is of relatively less important to voice and music transmission. But it can be very serious for video transmission

→ β is not a constant multiplied by (ω)(ang. freq) & therefore the line will introduced delay distortion

WKT,

$$P = \alpha + j\beta = \sqrt{ZY} = \sqrt{(R+j\omega L)(G+j\omega C)}$$

equate these 2 eqn

$$\alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)} \rightarrow (1)$$

$$(\alpha + j\beta)^2 = (R+j\omega L)(G+j\omega C)$$

$$\alpha^2 - \beta^2 + 2\alpha\beta j = (R+j\omega L)(G+j\omega C)$$

$$\alpha^2 - \beta^2 + j2\alpha\beta = RG + jR\omega C + j\omega LG - \omega^2 LC \rightarrow (2)$$

$$\alpha^2 - \beta^2 = RG - \omega^2 LC \quad (\text{Real}) \rightarrow (3)$$

$$2\alpha\beta = R\omega C + \omega LG \quad (\text{img}) \rightarrow (4)$$

Square on eqn (4).

$$4\alpha^2\beta^2 = \omega^2 (RC + LG)^2$$

$$\beta^2 = \frac{\omega^2 (RC+LG)^2}{4\alpha^2} \rightarrow (5)$$

Sub on eqn (3).

$$\cancel{\frac{\omega^2 (RC+LG)^2}{4\beta^2} - \beta^2 = RG - \omega^2 LC}$$

$$\cancel{\omega^2 (RC+LG)^2 - 4\beta^4 = RG - \omega^2 LC}$$

$$\alpha^2 - \frac{\omega^2 (RC+LG)^2}{4\alpha^2} = RG - \omega^2 LC$$

$$4\alpha^4 - \omega^2 (RC+LG)^2 = 4\alpha^2 (RG - \omega^2 LC)$$

$$4\alpha^4 - 4\alpha^2 (RG - \omega^2 LC) - \omega^2 (RC+LG)^2 = 0$$

$(\div 4)$ $$\alpha^4 - \alpha^2 (RG - \omega^2 LC) - \frac{\omega^2}{4} (RC+LG)^2 = 0$$

$x = \alpha$

$$x^2 - x(RG - \omega^2 LC) - \frac{\omega^2}{4}(RC + LG)^2 = 0$$

$ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

$a = 1, \quad c = -\frac{\omega^2}{4}(RC + LG)^2$

$b = -(RG - \omega^2 LC)$

$$x = \frac{(RG - \omega^2 LC) \pm \sqrt{(RG - \omega^2 LC)^2 + 4(1)\frac{\omega^2}{4}(RC + LG)^2}}{2}$$

$$= \frac{RG - \omega^2 LC \pm \sqrt{(RG - \omega^2 LC)^2 + \omega^2 (RC + LG)^2}}{2}$$

$x = \alpha^2$

$\alpha = \sqrt{x}.$

$$\alpha = \sqrt{\frac{RG - \omega^2 LC \pm \sqrt{(RG - \omega^2 LC)^2 + \omega^2 (RC + LG)^2}}{2}}$$

$$\cancel{\alpha = \frac{\sqrt{RG - \omega^2 LC} \pm (RG - \omega^2 LC)^2 + \omega^2 (RC + LG)^2}{\sqrt{2}}}$$

lly,

$$\beta = \sqrt{\left\{\frac{-RG + \omega^2 LC \pm \sqrt{(RG - \omega^2 LC)^2 + \omega^2 (RC + LG)^2}}{2}\right\}}$$

distortion | distortion less

$\alpha \rightarrow$ purely depe

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Distortion less line | dissipation less

$\alpha \rightarrow$ Independent of frequency

$\beta \rightarrow$ Direct function of frequency

WKT,

make the

$$\alpha = \sqrt{\left\{\frac{(RG-\omega^2 LC) \pm \sqrt{(RG-\omega^2 LC)^2 + \omega^2 (RC+LG)^2}}{2}\right\}}$$

to make the eqn as freq independent

Assume

$$(RG-\omega^2 LC)^2 + \omega^2 (RC+LG)^2 = (RG+\omega^2 LC)^2$$

$$\alpha = \sqrt{\left\{\frac{(RG-\omega^2 LC) + (RG+\omega^2 LC)}{2}\right\}}$$

$$\alpha = \sqrt{\frac{2RG}{2}}$$

$$\boxed{\alpha = \sqrt{RG}}$$ $\rightarrow$ independent of frequency.

pure

$Z = R + j\omega L$ $\rightarrow$ img (freq depend)

$Y = G + j\omega C$

lly,

$$\beta = \frac{(\omega^2 LC - RG) \pm RG + \omega^2 LC}{2}$$

$$\beta = \sqrt{\omega^2 LC}$$

$$\boxed{\beta = \omega\sqrt{LC}}$$ $\rightarrow$ purely dependent of frequency.

The condition of distortion less line

$$(RG-\omega^2 LC)^2 + \omega^2 (RC+LG)^2 = (RG+\omega^2 LC)^2$$

$RG - \omega^2 LC + \omega RC + \omega LG - RG - \omega^2 LC = 0$

$(-2\omega^2 LC + \omega RC + \omega LG)$ $\quad$ $\omega[-2\omega LC + RC + LG]$

$$RG - \omega^2 LC + \omega(RC+LG) = RG + \omega^2 LC$$

$$RG - \omega^2 LC + \omega RC + \omega LG = RG + \omega^2 LC$$

(or)

$$-2\omega^2 LC + \omega RC + \omega LG$$

$$\omega [-2\omega LC + RC + LG]$$

$$R^2G^2 + \omega^4 L^2C^2 - 2RG\omega^2 LC + \omega^2 [R^2C^2 + L^2G^2 + 2RCLG] = R^2G^2 + \omega^4 L^2C^2 + 2RG\omega^2 LC$$

$$\cancel{R^2G^2} + \cancel{\omega^4 L^2C^2} - 2RG\omega^2 LC + \omega^2 R^2C^2 + \omega^2 L^2G^2 + 2\omega^2 RCLG = \cancel{R^2G^2} + \cancel{\omega^4 L^2C^2} + 2RG\omega^2 LC$$

$$-4RG\omega^2 LC + \omega^2 R^2C^2 + \omega^2 L^2G^2 + 2\omega^2 RCLG = 0.$$

$$\omega^2 [-4RGLC + R^2C^2 + L^2G^2 + 2RCLG] = 0$$

$$\omega^2 [-2RGLC + R^2C^2 + L^2G^2] = 0$$

$$\omega^2 [(RC - LG)^2] = 0$$

$$RC - LG = 0$$

$$\boxed{RC = LG}$$

$$\boxed{\frac{R}{G} = \frac{L}{C}}$$

$$Z = R + j\omega L \qquad Y = G + j\omega C$$

$$Z_0 = \sqrt{\frac{Z}{Y}} \qquad Z = L\left[\frac{R}{L} + j\omega\right] \qquad Y = C\left[\frac{G}{C} + j\omega\right]$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{R(1 + j\omega L/R)}{G(1 + j\omega C/G)}} \quad \text{(or)} \quad \sqrt{\frac{L(R/L + j\omega)}{C(G/C + j\omega)}}$$

wkt,

$$RC = LG$$

$$\frac{L}{R} = \frac{C}{G}$$

$$Z_0 = \sqrt{\frac{R(1 + j\omega L/R)}{G(1 + j\omega L/R)}}$$

$$\boxed{Z_0 = \sqrt{\frac{R}{G}}}$$

$$RC = LG$$

$$\frac{R}{L} = \frac{G}{C}$$

$$Z_0 = \sqrt{\frac{L(R/L + j\omega)}{C(R/L + j\omega)}}$$

$$\boxed{Z_0 \sqrt{\frac{L}{C}}}$$